

Assignment 3 :

CCAS.4.3

Name : Adham Anbar

ID : 202202760

Starting with tables inserted as *provided* (Airline Reservation)

AIRPLANE [-]
Airplane_id [text]
Total_number_of_seats [integer]
Airplane_type [text]

AIRPLANE_TYPE [-]
Airplane_type_name [text]
Max_seats [integer]
Company [text]

AIRPORT [-]
Airport_code [text]
Name [text]
City [text]
State [text]

CAN_LAND [-]
Airplane_type_name [text]
Airport_code [text]

FARE [-]
Flight_number [text]
Fare_code [text]
Amount [real]
Restrictions [text]

FLIGHT [-]
Flight_number [text]
Airline [text]
Weekdays [text]

FLIGHT_LEG [-]
Flight_number [text]
Leg_number [integer]
Departure_airport_code [text]
Arrival_airport_code [text]
Scheduled_departure_time [datetime]
Scheduled_arrival_time [datetime]

SEAT_RESERVATION [-]
Flight_number [text]
Leg_number [integer]
Date [date]
Seat_number [text]
Customer_name [text]
Customer_phone [text]

Bonus

Available Tables

FLIGHT_LEG

Flight_number	Leg_number	Departure_airport_code	Arrival_airport_code
F101	1	CAI	LHR

LEG_INSTANCE

Flight_number	Leg_number	Date	Number_of_available_seats	Airplane
F101	1	2025-11-10	160	A100

Available Tables

AIRPLANE

Airplane_id	Total_number_of_seats	Airplane_type
A100	180	E320JET

AIRPLANE_TYPE

Airplane_type_name	Max_seats	Company
E320JET	180	Airbus

AIRPORT

Airport_code	Name	City	State
CAI	Cairo Intl	Cairo	EG
LHR	Heathrow	London	UK

Available Tables

FARE

Flight_number	Fare_code	Amount	Restrictions
F101	ECON	100	Refundable

FLIGHT

Flight_number	Airline	Weekdays
F101	EgyptAir	Monday
F202	British Air	Tuesday

FLIGHT_LEG

Flight_number	Leg_number	Departure_airport_code	Arrival_airport_code
F101	1	CAI	LHR

```
INSERT INTO AIRPORT VALUES ('CAI', 'Cairo Intl', 'Cairo', 'EG');
INSERT INTO AIRPORT VALUES ('LHR', 'Heathrow', 'London', 'UK');
```

```
INSERT INTO FLIGHT VALUES ('F101', 'EgyptAir', 'Monday');
INSERT INTO FLIGHT VALUES ('F202', 'British Air', 'Tuesday');
```

```
INSERT INTO FLIGHT_LEG VALUES ('F101', 1, 'CAI', 'LHR', '2025-11-10 08:00:00', '2025-11-10 12:00:00');
```

Ok, so I used programiz (online complier) then I created all tables required for the database schema and my references is from (LAB4:the fruits one)

Question A

```
-- Question A
SELECT
    Flight_number,
    Airline,
    Weekdays
FROM
    FLIGHT
ORDER BY
    Airline ASC;

-- we try inserting some random values so that the tables doesnot seem empty
INSERT INTO AIRPORT VALUES ('CAI', 'Cairo Intl', 'Cairo', 'EG');
INSERT INTO AIRPORT VALUES ('LHR', 'Heathrow', 'London', 'UK');

INSERT INTO FLIGHT VALUES ('F101', 'EgyptAir', 'Monday');
```

Output

Flight_number	Airline	Departure_airport_code	Departure_Airport	Arrival_airport_code	Arrival_Airport
F101	EgyptAir	CAI	Cairo Intl	LHR	Heathrow

I inserted random values because there was no output given

Question B

```

/*-- Question B

SELECT
    Flight_number,
    Restrictions,
    Amount AS Original_Amount,
    (Amount * 1.15) AS Amount_After_Tax
FROM
    FARE
WHERE
    Amount > 50;*/

```

Output

Available Tables

Flight_number	Restrictions	Original_Amount	Amount_After_Tax
F101	Refundable	100	114.99999999999999

Question D

```

-- Question D
SELECT
    AT.Airplane_type_name,
    A.Total_number_of_seats
FROM
    AIRPLANE_TYPE AT
JOIN
    AIRPLANE A
ON
    AT.Airplane_type_name = A.Airplane_type;

```

Output

Available Tables

Airplane_type_name	Total_number_of_seats
E320JET	180

Question E

```
-- Question E
SELECT
    Airplane_type_name
FROM
    AIRPLANE_TYPE
WHERE
    Airplane_type_name LIKE 'E%T';
```

Output

Available Tables

Airplane_type_name
E320JET

Question F

```
-- Question F
SELECT
    F.Flight_number,
    F.Airline,
    FL.Departure_airport_code,
    DA.Name AS Departure_Airport,
    FL.Arrival_airport_code,
    AA.Name AS Arrival_Airport,
    FL.Scheduled_departure_time,
    FL.Scheduled_arrival_time,
    LI.Airplane_id,
    AT.Airplane_type_name,
    'Monday' AS Weekday
FROM
    FLIGHT F
JOIN
    FLIGHT_LEG FL ON F.Flight_number = FL.Flight_number
JOIN
    LEG_INSTANCE LI ON FL.Flight_number = LI.Flight_number
JOIN
    AIRPLANE A ON LI.Airplane_id = A.Airplane_id
JOIN
    AIRPLANE_TYPE AT ON A.Airplane_type = AT.Airplane_type_name
JOIN
    AIRPORT DA ON FL.Departure_airport_code = DA.Airport_code
JOIN
    AIRPORT AA ON FL.Arrival_airport_code = AA.Airport_code
WHERE
    F.Weekdays = 'Monday';
```

Output

Flight_number	Airline	Departure_airport_code	Departure_Airport	Arrival_airport_code	Arrival_Airport	Scheduled_departure_time	Scheduled_arrival_time	Airplane_id	Airplane_type_name	Weekday
F101	EgyptAir	CAI	Cairo Intl	LHR	Heathrow	2025-11-10 08:00	2025-11-10 12:00	A100	E320JET	Monday

Question J

```
-- Question J
SELECT
    F.Flight_number,
    F.Airline,
    DA.Name AS Departure_Airport,
    AA.Name AS Arrival_Airport,
    FL.Scheduled_departure_time,
    FL.Scheduled_arrival_time
FROM
    FLIGHT F
JOIN
    FLIGHT_LEG FL ON F.Flight_number = FL.Flight_number
JOIN
    AIRPORT DA ON FL.Departure_airport_code = DA.Airport_code
JOIN
    AIRPORT AA ON FL.Arrival_airport_code = AA.Airport_code
WHERE
    F.Airline LIKE '%Air%';
```

Output

Flight_number	Airline	Departure_airport_code	Departure_Airport	Arrival_airport_code	Arrival_Airport	Scheduled_departure_time	Scheduled_arrival_time	Airplane_id	Airplane_type
F101	EgyptAir	CAI	Cairo Intl	LHR	Heathrow	2025-11-10 08:00	2025-11-10 12:00	A100	E320JET

Flight_number	Airline	Departure_Airport	Arrival_Airport	Scheduled_departure_time	Scheduled_arrival_time
F101	EgyptAir	Cairo Intl	Heathrow	2025-11-10 08:00	2025-11-10 12:00